

# Animesh Sharma

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## EDUCATION

### Parul University

B.Tech in Computer Science & Engineering | CGPA: 7.99

Vadodara, Gujarat

2023 – 2027

## EXPERIENCE

### Open Source Contributor

May 2026

GirlScript Summer of Code (GSSoC) — CommitPulse (GitHub analytics platform)

- Fixed unbounded **TTLCache** OOM leak (`lib/cache.ts`): added 60s active GC sweep, FIFO `maxSize` eviction, `timer.unref()` process protection, and a `destroy()` lifecycle; covered by 8 unit tests.
- Replaced `Promise.all` with `Promise.allSettled` in the GitHub API fetch layer; added `AbortSignal` for hung requests and typed fallbacks, preventing a single rate-limit from crashing the full dashboard.

### Freelance Software Engineer

Feb 2026

BiteBox — Cafe Billing & POS System | Flutter, Firebase, Dart

- Developed a custom Flutter POS app that **eliminated 100% of recurring SaaS costs** (Rs. 12,000/yr) by leveraging the Firebase free-tier.
- Architected a **dual-database system** (SQLite + Firebase) for offline-first capabilities, atomic multi-device cart sync, and **sub-50ms** analytics.
- Engineered atomic invoice generation, ESC/POS Bluetooth thermal printing, and secure role-based access control.

## PROJECTS

### Parallax — Collaborative Development Platform | React, TypeScript, Spring Boot, WebRTC, Docker

[GitHub]

- Full-stack collaborative IDE supporting **50+ concurrent users** with OT-inspired sync, live cursors, and multi-file editing via Monaco Editor.
- **WebRTC** peer-to-peer voice/video using a custom STOMP/WebSocket signaling engine, integrated directly in-IDE.
- **Docker-orchestrated** isolated code execution sandbox achieving **sub-150ms** cold-start via warm container pre-allocation.

### NextLevel — Productivity Platform | Spring Boot, Spring AI, MongoDB Atlas Vector Search

[Live] [GitHub]

- Semantic knowledge vault using **Spring AI**, **Groq Llama-3.3**, and MongoDB Atlas Vector Search (768-dim embeddings) with cosine similarity retrieval beyond keyword matching.
- Fault-tolerant AI ingestion pipeline via **JobRunr** + **Resilience4j** (exponential backoff) — asynchronously scrapes, summarizes, and vectorizes content with automatic LLM failure recovery.
- Real-time **RAG** assistant with SSE streaming, a custom **SM-2 spaced repetition engine**, and gamification layer for personalized learning.

### JSON Parser Library | Java, JUnit 5

[GitHub]

- **Zero-dependency** Java library with a custom lexer and recursive descent parser for parsing, querying, validating, and diffing JSON schemas.
- Event-driven streaming parser using a custom state machine — processes **10MB+** nested payloads in **O(1)** space, reducing peak heap by **>95%**.
- **JSONPath-like** query syntax with O(1) deep-node indexing and column-level syntax error diagnostics.

## TECHNICAL SKILLS

**Languages:** Java, JavaScript, TypeScript, Python, Dart, C, SQL

**Backend:** Spring Boot, Spring MVC, Spring Security, Hibernate, JPA, Apache Kafka, Redis, REST APIs, JWT, WebSockets

**Frontend:** React, Vite, Tailwind CSS, WebRTC, Monaco Editor

**AI/ML:** Spring AI, PgVector, LLM Integration (OpenAI API), Machine Learning

**Cloud / DevOps:** Docker, AWS, Firebase, Git

**Databases:** PostgreSQL, MongoDB, MySQL, SQLite

## CERTIFICATIONS & ACHIEVEMENTS

- ServiceNow Certified Application Developer (**CAD**).
- ServiceNow Certified System Administrator (**CSA**).
- HackerRank Certified: Java & Python.
- Selected for **State-Level Regional** at OpenAI Academy × NxtWave × IndiaAI Hackathon.
- Solved **700+ DSA problems**.